Experiment 9.1

Simulation of Ground Wave and Sky Wave Propagation using MATLAB

Aim

To simulate ground wave and sky wave propagation and to calculate the field strength, skip distance and maximum usable frequency (MUF).

Software Required

MATLAB (MATLAB Online)

Procedure / Algorithm

- Take transmitted power Pt = 10 kW and distance d = 1 to 500 km; compute the ground wave field strength.
- Take ionosphere height h = 250 km, angle of incidence = 60° and critical frequency fc = 5 MHz.
- Calculate skip distance = $2h \cdot \tan(\theta)$ and MUF = $fc \cdot \sec(\theta)$.
- Plot the ground wave, MUF, skip distance and ground wave vs sky wave comparison.

Output

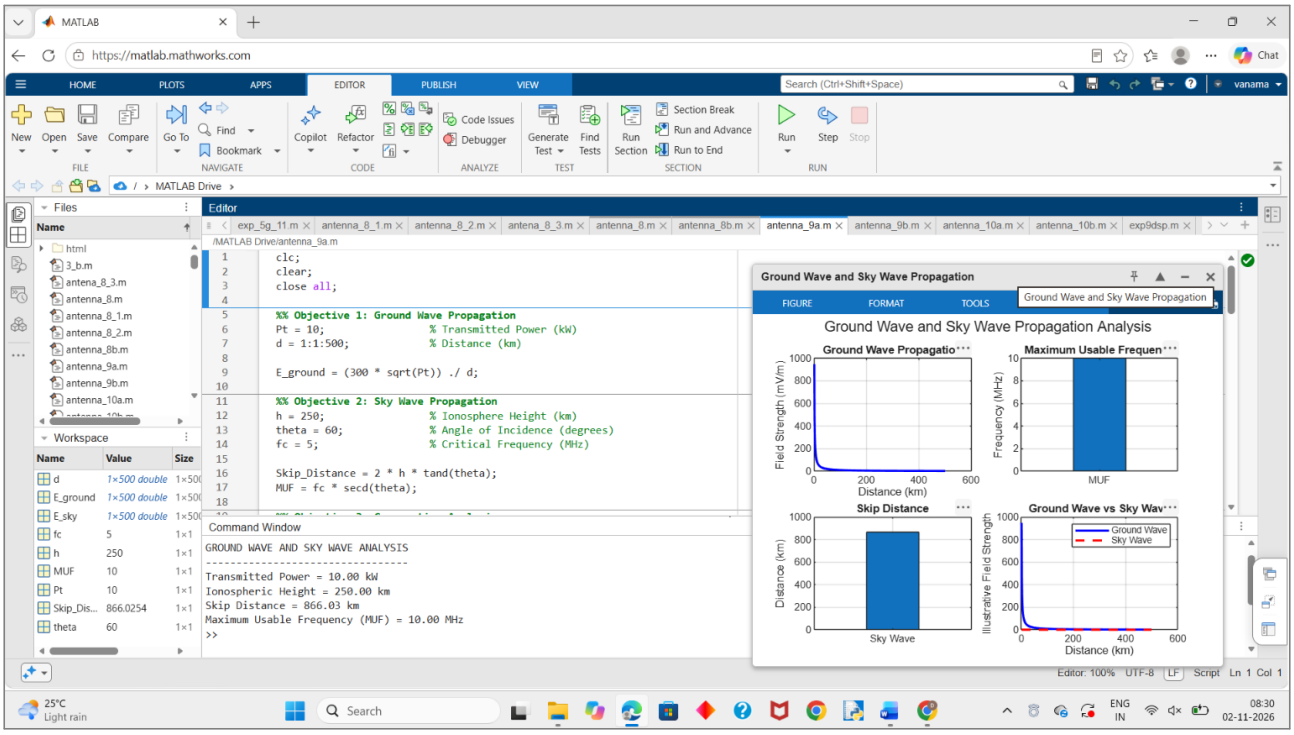


Fig. 9.1 Ground wave and sky wave propagation analysis

Result

- Skip distance = 866.03 km and MUF = 10 MHz.
- Ground wave field strength falls quickly with distance, so sky wave propagation is needed for long-distance communication.